练习：打球决策树

outlook	temperature	humidity	windy	play
sunny	hot	high	FALSE	no
sunny	hot	high	TRUE	no
overcast	hot	high	FALSE	yes
rainy	mild	high	FALSE	yes
rainy	cool	normal	FALSE	yes
rainy	cool	normal	TRUE	no
overcast	cool	normal	TRUE	yes
sunny	mild	high	FALSE	no
sunny	cool	normal	FALSE	yes
rainy	mild	normal	FALSE	yes
sunny	mild	normal	TRUE	yes
overcast	mild	high	TRUE	yes
overcast	hot	normal	FALSE	yes
rainy	mild	high	TRUE	no

数据：

14天的气象数据(属性：outlook，temperature，humidity，windy)，并已知这些天气是否打球(play)。

问题：

根据决策树算法，构建一棵是否打球的决策树。并给出每个结点选择分裂属性时所作的计算。

解：

① 总数据集记为 D，

系统整体信息熵：$Ent(D) = -\sum_{k=1}^{|y|} p_k \cdot \log_2(p_k)$.

代入数据：$Ent(D) = -\frac{9}{14}\log_2(\frac{9}{14}) - \frac{5}{14}\log_2(\frac{5}{14}) \approx 0.940$.

② 计算4个属性的信息增益：

1° outlook(天气)属性.

$Ent(D_{sunny}) = -\frac{2}{5}\log_2(\frac{2}{5}) - \frac{3}{5}\log_2(\frac{3}{5}) \approx 0.971$.

$Ent(D_{overcast}) = -\frac{4}{4}\log_2(\frac{4}{4}) - 0 = 0$.

$Ent(D_{rainy}) = -\frac{3}{5}\log_2(\frac{3}{5}) - \frac{2}{5}\log_2(\frac{2}{5}) \approx 0.971$

$Gain(D, outlook) = 0.940 - (\frac{5}{14}\times 0.971 + \frac{4}{14}\times 0 + \frac{5}{14}\times 0.971) \approx 0.247$.

2° temperature 属性.

$Ent(D_{hot}) = -\frac{2}{4}\log_2(\frac{2}{4}) - \frac{2}{4}\log_2(\frac{2}{4}) = 1.0$

$Ent(D_{mild}) = -\frac{4}{6}\log_2(\frac{4}{6}) - \frac{2}{6}\log_2(\frac{2}{6}) \approx 0.918$

$Ent(D_{cool}) = -\frac{3}{4}\log_2(\frac{3}{4}) - \frac{1}{4}\log_2(\frac{1}{4}) \approx 0.811$

$Gain(D, temperature) = 0.940 - (\frac{4}{14}\times 1.0 + \frac{6}{14}\times 0.918 + \frac{4}{14}\times 0.811) \approx 0.029$

3° humidity 属性

$Ent(D_{high}) = -\frac{3}{7}\log_2(\frac{3}{7}) - \frac{4}{7}\log_2(\frac{4}{7}) \approx 0.985$.

$Ent(D_{normal}) = -\frac{6}{7}\log_2(\frac{6}{7}) - \frac{1}{7}\log_2(\frac{1}{7}) \approx 0.592$.

$Gain(D, humidity) = 0.940 - (\frac{7}{14}\times 0.985 + \frac{7}{14}\times 0.592) \approx 0.151$.

4° windy 属性

$Ent(D_{FALSE}) = -\frac{6}{8}\log_2(\frac{6}{8}) - \frac{2}{8}\log_2(\frac{2}{8}) = 0.811$

$Ent(D_{TRUE}) = -\frac{3}{6}\log_2(\frac{3}{6}) - \frac{3}{6}\log_2(\frac{3}{6}) = 1.0$

$Gain(D, windy) = 0.940 - (\frac{8}{14} \times 0.811 + \frac{6}{14} \times 1.0) \approx 0.048.$

∵ Gain(D, outlook) max

∴ 选择 outlook 作为根节点.

③ 构建子节点.

1° 分支 outlook = sunny.

$Ent(D_{sunny}, hot) = 0$

$Ent(D_{sunny}, mild) = -\frac{1}{2}\log_2(\frac{1}{2}) - \frac{1}{2}\log_2(\frac{1}{2}) = 1.0$

$Ent(D_{sunny}, cool) = 0.$

$Gain(D_{sunny}, temp) = Ent(D_{sunny}) - (\frac{2}{5} \times 0 + \frac{2}{5} \times 1.0 + \frac{1}{5} \times 0) = 0.571.$

同理, $Gain(D_{sunny}, humidity) = 0.971.$

$Gain(D_{sunny}, windy) = 0.020.$

⇓

选择 humidity 作为划分属性

2° 分支 outlook = rainy.

同理, 选择 windy 作为划分属性

3° 分支 outlook = overcast

⇒ 叶子节点全部为 yes

